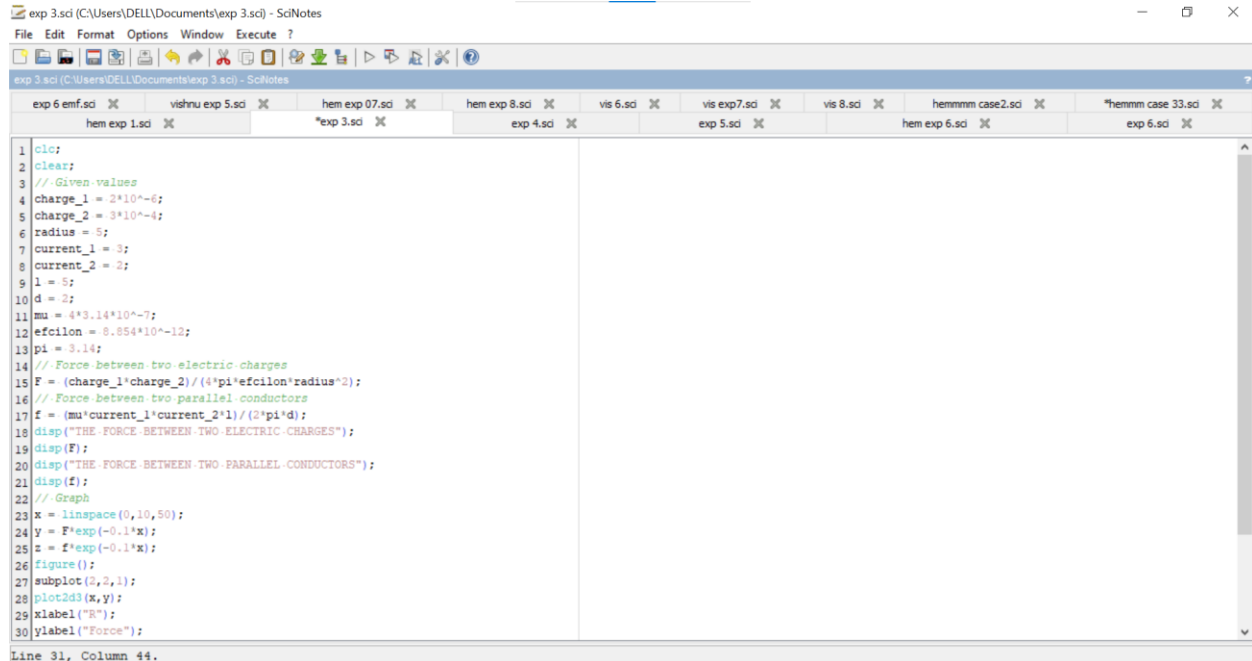
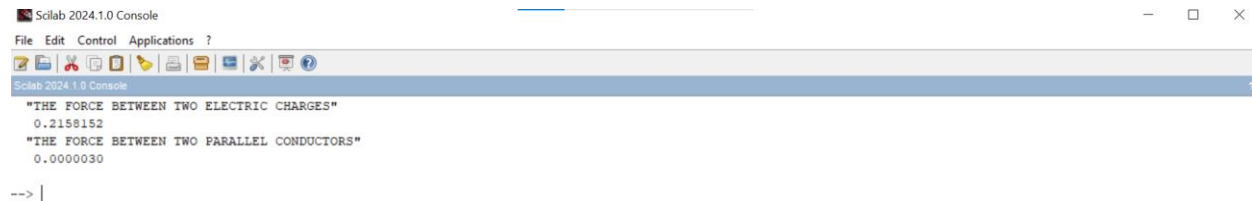


EXP 3: Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.



```
1 clc;
2 clear;
3 // Given values
4 charge_1 = 2*10^-6;
5 charge_2 = 3*10^-4;
6 radius = 5;
7 current_1 = 3;
8 current_2 = 2;
9 l = 5;
10 d = 2;
11 mu = 4*3.14*10^-7;
12 epsilon = 8.854*10^-12;
13 pi = 3.14;
14 // Force between two electric charges
15 F = (charge_1*charge_2)/(4*pi*epsilon*radius^2);
16 // Force between two parallel conductors
17 f = (mu*current_1*current_2*l)/(2*pi*d);
18 disp('THE FORCE BETWEEN TWO ELECTRIC CHARGES');
19 disp(F);
20 disp('THE FORCE BETWEEN TWO PARALLEL CONDUCTORS');
21 disp(f);
22 // Graph
23 x = linspace(0,10,50);
24 y = F*exp(-0.1*x);
25 z = f*exp(-0.1*x);
26 figure();
27 subplot(2,2,1);
28 plot(2d3(x,y));
29 xlabel('R');
30 ylabel('Force');
```

Line 31, Column 44.



```
Scilab 2024.1.0 Console
File Edit Control Applications ?

"THE FORCE BETWEEN TWO ELECTRIC CHARGES"
0.2158152
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"
0.0000030

--> |
```

d

Exp 36 - Determination and plotting of force of attraction
btw two electric charges and two electric charges and
two parallel conductors separated by varying distance.

Charge 1: 2×10^{-6}

Charge 2: 3×10^{-4}

$r = 5 \text{ mm}$

Current 1 = 3

Current 2 = 2

$l = 5 \text{ mm}$

$D = 2 \text{ mm}$

Sol

$$F = \frac{(2 \times 10^{-6})(3 \times 10^{-4})}{4\pi(8.854 \times 10^{-12})(5 \times 10^{-3})^2}$$

$$= \frac{(6 \times 10^{-10})}{(2.781 \times 10^{-15})}$$

$F \approx 215700 \text{ N}$

0/pv

$$f = \frac{(4\pi \times 10^{-7})(3)(2)(5)}{2\pi(2 \times 10^{-3})}$$

$$f = \frac{120 \times 10^{-7}}{4 \times 10^{-3}}$$

$$= 3 \times 10^{-3} \text{ N}$$

$f = 0.003 \text{ N}$

